Data Analysis and Tax Compliance

R laboratory
Sampling means and bivariate analysis

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Two questions for the R laboratory

- How does the sample mean vary across repeated samples? Municipal pension income, with sample sizes 10, 100 and 1,000.
- How do we analyse a relationship between two variables? BES education indicators, tables, groups and quantitative pairs.

Open the laboratory guide and download the complete R project

The RStudio project

Extract the project and open `HCAI_R_Lab.Rproj`.

```
getwd()
list.files("data")
source("scripts/01_sampling_mean.R")
source("scripts/02_bivariate.R")
```

Run selected lines with Ctrl+Enter, or Cmd+Enter on a Mac. Both scripts use base R. Results appear in `results/`.

Part A: sampling distributions

The observations vary across municipalities.

The estimate varies across samples.

Revisit the inference e-tivity

The municipal income file

The supplied teaching snapshot contains **8,047 municipalities**.

```
municipalities <- read.csv("data/redditi.csv",
  check.names = FALSE,
  colClasses = c(Cod = "character"))
nrow(municipalities)
head(municipalities)
```

`Rp_avg`: municipal pension-income average, in EUR.
`Rp_share`: pension income / taxable income, as a percentage.

The target determines the mean

```
mean(municipalities$Rp_avg)
# 14366.97 EUR
sum(municipalities$Rp_tot) /
  sum(municipalities$Rp_freq)
# 16703.62 EUR
```

The first gives each municipality the same weight. The second weights municipal means by pension-income frequency.

Discussion

Which quantity describes the typical municipality? Which uses the aggregate amount and frequency?

One random sample

```
set.seed(1111)
N <- nrow(municipalities)
n <- 100
selected <- sample.int(N, n, replace = FALSE)
one_sample <- municipalities[selected, ]
mean(one_sample$Rp_avg)
# 14309.60 EUR
```

Sampling error: $14,309.60 - 14,366.97 = -57.37$ EUR. The seed reproduces this particular draw.

Sample size and repetitions

$N = 8047$	Municipalities in the finite teaching population
$n = 10, 100, 1000$	Municipalities in one sample
$B = 200$	Independent repetitions of the complete sampling experiment

A municipality appears at most once within a sample. It can reappear in another replicate.

Discussion

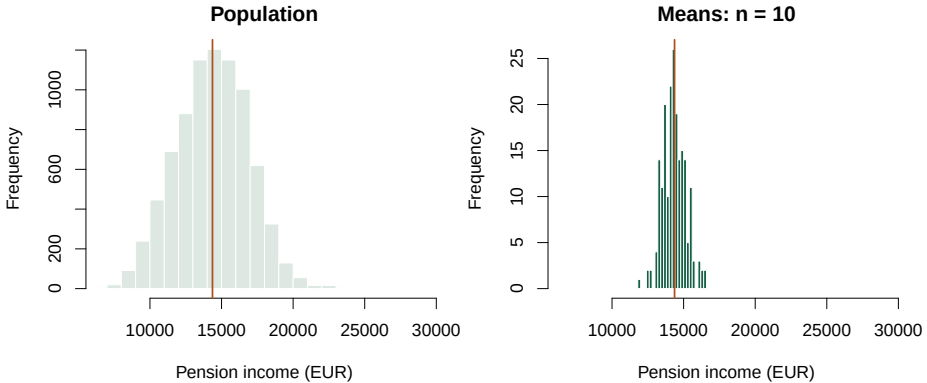
Increasing B describes the sampling distribution more accurately. Does it change the uncertainty of one sample of size n?

Repeated samples in R

```
set.seed(20261111)
B <- 200
n <- 10
means_n10 <- replicate(B, {
  selected <- sample.int(N, n, replace = FALSE)
  mean(municipalities$Rp_avg[selected])
})
mean(means_n10)
sd(means_n10)
```

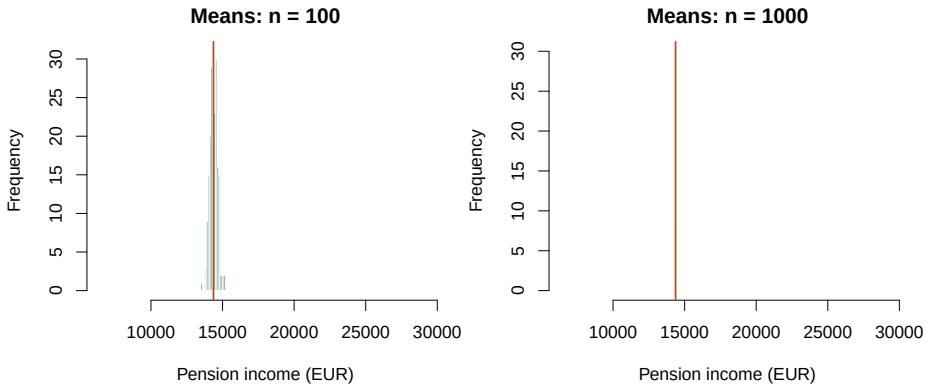
The supplied script repeats the calculation for both variables and all three sample sizes.

Population and means from small samples



Both panels use the same horizontal range. The right panel contains means of whole samples.

Means from larger samples



The scale is unchanged. Larger samples make the distribution of the mean more concentrated.

The standard error under this design

Use the finite-population variance with denominator $N - 1$:

$$S^2 = \frac{1}{N-1} \sum_{i=1}^{N} (y_i - \mu)^2.$$

For simple random sampling without replacement,

$$\mathrm{SE}(\bar{Y}) = \sqrt{\frac{1 - n/N}{n} S^2}.$$

Larger n reduces the spread of the estimator. At $n = N$, the supplied population mean is known exactly.

Empirical and theoretical standard errors

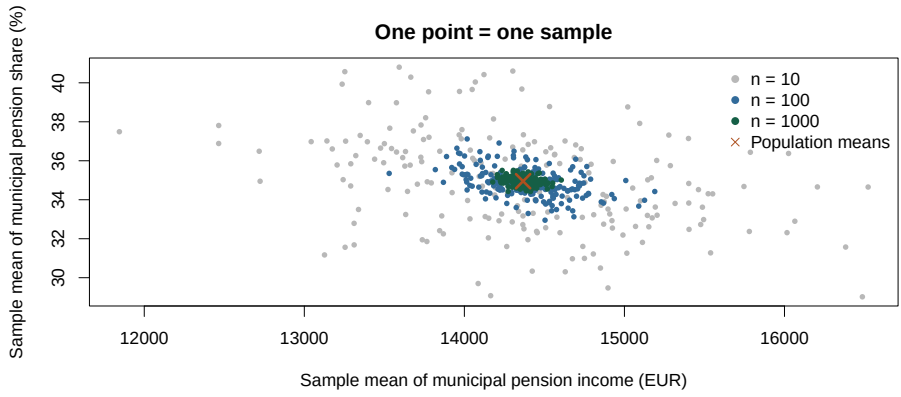
n	Mean of 200 means	Empirical SE	Theoretical SE
10	14,347.04	788.94	800.59
100	14,384.03	264.75	251.75
1,000	14,367.89	72.64	74.97

All values are in EUR. The target is $14,366.97$ EUR.

Discussion

Why do empirical and theoretical values differ? What would increasing B change?

Two sample means at once



Each point summarises one sample. Larger samples concentrate near the pair of population means.

A restricted sampling frame

An optional simulation samples 100 municipalities only from the highest-income quarter.

Target mean, all supplied municipalities	14,366.97 EUR
Mean of the restricted-frame estimates	17,548.66 EUR
SD of the restricted-frame estimates	127.60 EUR

The procedure estimates a different frame mean with considerable precision.

Discussion

How could we change the design to address the original target?

Part A: interpretation exercise

Change the number of repetitions to 2,000, keeping the sample sizes fixed.

- Compare the means and empirical standard errors with the reference table.
- Explain the different roles of n and B.
- State the unit of analysis and target population.
- Explain why the population observations need not become normally distributed.

Part B: bivariate analysis

The question and the two variable types determine the method.

Revisit the bivariate e-tivity

The BES education file

```
bes <- read.csv("data/bes_education.csv")
nrow(bes)
# 20
names(bes)
table(bes$area)
```

One row represents a region. We retain Trentino-Alto Adige and exclude its two autonomous provinces to avoid overlapping observations.

Questions and methods

Question	Starting point
Area and high early leaving	Contingency table, conditional proportions, association
NEET across two dropout groups	Group distributions and Welch's t test
NEET across four areas	Group distributions and one-way ANOVA
Culture and tertiary education	Scatterplot and Pearson correlation

A categorical variable from a threshold

```
bes$dropout <- factor(
  ifelse(bes$early_leaving_pct > 15,
         "Above 15%", "At most 15%"),
  levels = c("At most 15%", "Above 15%"))
tab <- table(bes$area, bes$dropout)
prop.table(tab, margin = 1) * 100
```

The threshold defines the comparison before the test. It discards differences within each category.

Counts and conditional percentages

Area	At most 15%	Above 15%	Total
North-West	4	0	4
North-East	4	0	4
Centre	4	0	4
South-Islands	4	4	8
Total	16	4	20

Within South-Islands, 50% of regions exceed the threshold. All four above-threshold regions belong to South-Islands.

Discussion

State the denominator in each sentence.

Expected counts before a p-value

```
chi <- suppressWarnings(chisq.test(tab,
                                   correct = FALSE))
chi$expected
sum(chi$expected < 5)
# 7 of 8 cells
fisher.test(tab)
# p-value = 0.07699
```

Here $\chi^2 = 7.5$ and Cramér's $V = 0.6124$. Sparse expected counts undermine the asymptotic approximation.

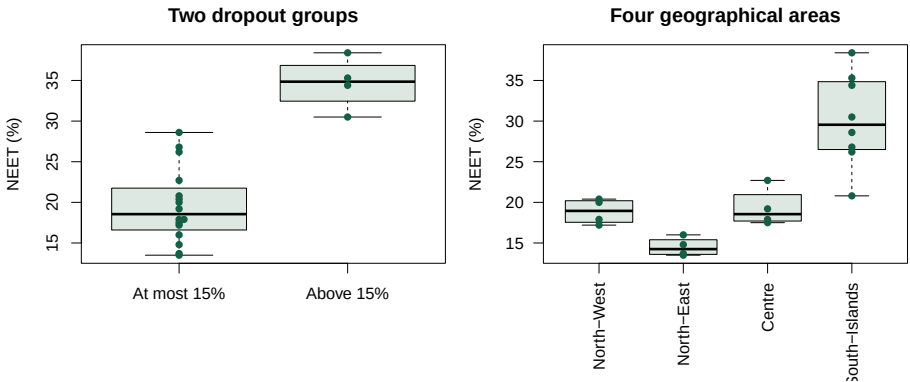
Exact tests and the inferential model

- Fisher's test conditions on the observed margins under its sampling model.
- It addresses sparse counts in this comparison.
- It does not establish independence of neighbouring regions or remove confounding.
- These 20 regions form the observed territorial set. Inference requires a model beyond describing that set.

Discussion

What can we state directly from the table before choosing an inferential model?

NEET percentages across groups



Each dot is one region. Mean NEET is 34.65% in the four above-threshold regions and 19.575% in the other sixteen.

Two groups: Welch's t test

```
t.test(neet_pct ~ dropout, data = bes)
```

Mean difference, group 1 minus group 2	−15.075 percentage points
t and degrees of freedom	−7.586, 6.367
95% interval for the difference	[−19.870, −10.280]
Two-sided model-based p-value	0.000205

Group order: *At most 15%* minus *Above 15%*. Inference relies on the comparison model and independent observations.

Several groups: one-way ANOVA

```
fit_anova <- aov(neet_pct ~ area, data = bes)
summary(fit_anova)
```

$$F(3, 16) = 16.856, \quad p = 0.0000330, \quad \eta^2 = 0.7596.$$

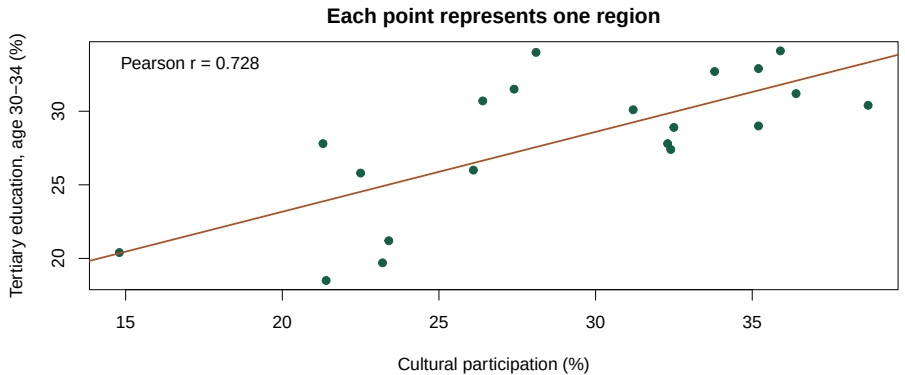
- The global alternative says at least one model mean differs.
- Classical ANOVA assumes independent, normal errors with a common variance.
- Inspect the small groups and consider unequal-variance methods.

Two quantitative variables

```
plot(tertiary_pct ~ cultural_pct, data = bes)
cor(bes$cultural_pct, bes$tertiary_pct)
# 0.7277
cor(bes[c("tertiary_pct", "cultural_pct",
          "reading_pct", "library_pct")])
```

The percentages describe regional aggregates. Correlation measures the linear relationship between those regional values.

Cultural participation and tertiary education



The positive regional association does not identify an individual effect or a causal mechanism.

Optional bridge: a fitted line

```
fit_line <- lm(tertiary_pct ~ cultural_pct,
               data = bes)
summary(fit_line)
```

$$\widehat{\text{tertiary percentage}} = 12.323 + 0.5424 \times \text{cultural percentage}.$$

$$R^2 = 0.5296.$$

A one-percentage-point difference in cultural participation is associated with about 0.542 percentage points in the fitted response. Predictive evaluation follows in supervised learning.

Part B: a short analytical report

- State the unit, variables and comparison.
- Report a descriptive effect in meaningful units.
- Explain the conditions behind any test you report.
- Distinguish regional association from a causal or individual claim.

Laboratory guide, scripts and data dictionary